

Revision Notes for Class 8 Maths Chapter 6 – We Distribute Yet Things Multiply

Algebra helps us write number patterns and relationships using letters and symbols. This chapter introduces the distributive property of multiplication over addition and shows how useful it is for simplifying expressions, finding products quickly, and solving different types of algebraic problems. By working with these properties, students can develop a deeper understanding of how multiplication interacts with addition.

Understanding Distributivity

The distributive property states that multiplying a number by a sum is the same as multiplying it by each term and adding the results. For example, $a \times (b + c) = ab + ac$. This property is the foundation for a lot of algebraic manipulation, like expanding brackets and solving equations.

When multiplying two numbers, let's say 23 and 27, it's helpful to see what happens if either number increases. If 23 is increased by 1, the increase in product is 27. If 27 is increased by 1, the increase in product is 23. If both are increased by 1, the product increases by $a + b + 1$. This can be written algebraically as $(a + 1)(b + 1) = ab + a + b + 1$.

General Changes in Multiplication

Multiplying $(a + m)(b + n)$ expands to $ab + an + bm + mn$, showing the total increase as $an + bm + mn$. Using this method makes it easier to solve complex multiplication problems, especially when both numbers are changed by different amounts. Visual models, like rectangle area diagrams, can make these calculations clearer.

When one number increases and the other decreases, for example $(a + 1)(b - 1)$, the product's change depends on the original numbers. Sometimes the product decreases. This shows that the net result depends on both the amount added and subtracted. Trying values like $a = 4, b = 2$ helps you see how the pattern works.

Expanding Expressions with Distributivity

Distributivity applies not just to numbers but to expressions too. For instance, the expansion $3a^2(a - b + 1/5)$ gives $3a^3 - 3a^2b + (3/5)a^2$. When expanding $(a + b)(a + b)$, the result is $a^2 + 2ab + b^2$. These outcomes, known as identities, are widely used in maths, especially when simplifying expressions and finding squares of binomials.

For more terms, distributivity still holds. For example, $(a + b)(a^2 + 2ab + b^2)$ expands to $a^3 + 3a^2b + 3ab^2 + b^3$. Recognizing and collecting “like terms” in such expansions is an important skill in algebra.

Shortcuts Using Distributive Property

The distributive property is very helpful for fast calculations. For instance, multiplying by 11 is easier if you write $N \times 11$ as $N \times (10 + 1)$, which equals $10N + N$. This trick works for any number of digits and can be extended to numbers like 101 and 1001. By breaking up numbers into easily multiplied parts, calculations can be done mentally with much less effort.

Special Algebraic Identities

Some common algebraic identities derived from distributivity include:

- $(a + b)^2 = a^2 + 2ab + b^2$
- $(a - b)^2 = a^2 - 2ab + b^2$
- $(a + b)(a - b) = a^2 - b^2$

Using these identities makes squaring or multiplying binomials much quicker. For example, to find the square of 65, use $(60 + 5)^2 = 3600 + 2 \times 60 \times 5 + 25 = 4225$.

Sometimes two numbers can be changed so that their product is not affected, such as increasing one by 2 and decreasing the other by 4, as in $(a + 2)(b - 4)$. By choosing values carefully, you can create many such examples, which encourages logical thinking.

Learning from Mistakes

It is easy to make errors when simplifying algebraic expressions. For example, $-3p(-5p + 2q)$ should be expanded using distributivity, giving $15p^2 - 6pq$.

Double-checking each step helps prevent common errors like sign mistakes or missing multiplications.

Information is also shared about patterns, such as tile arrangements or growing sequences (like circles or coins), which can be expressed algebraically. For instance, the number of circles in a pattern at step k can be written as $k^2 + 2k$, using reasoning and algebra to find general solutions.

Visualizing Area and Patterns

The distributive property is useful for calculating areas using different methods. For instance, the area of shaded tiles can be found by different approaches, such as $(m + n)^2 - 4mn$ or $(n - m)^2$. Different approaches can arrive at the same answer, demonstrating the flexibility of algebra.

Historical Connections

The distributive property has a rich history. Ancient mathematicians, like Brahmagupta, described “multiplication by parts” many centuries ago. Concepts we use now have been part of mathematical thinking across cultures and times.

Summary of Key Points

- Distributive property helps in breaking down and expanding products across addition and subtraction, making calculations and algebraic simplification easier.
- Special identities like $(a + b)^2$, $(a - b)^2$, and $(a + b)(a - b)$ are important tools for quick calculations.
- Visual models like rectangle or square areas can support understanding of algebraic patterns.
- There are often multiple methods to solve an algebraic problem, giving more flexibility in approach.
- Checking work carefully helps avoid common errors in sign, grouping, or multiplying.